📊 AI/ML Engineering Squad: Progress Metrics

- **Current Progress: 5%** (Commencing Week 1, Day 2 of the 3-Month Bootcamp).
- **Current Focus:** Week 1 Mandate — Data Engineering, Feature Engineering, and Exploratory Data Analysis (EDA).

📝 The Daily Drop Blueprint

Title: Deep-Tech & AI Squad: Week 1, Tuesday - Feature Engineering & Exploratory Data Analysis (EDA)

📂 Section A: Today's Materials (The Synthesis Mandate)

You must synthesize the information from all sources below to complete today's execution mandate. Do not consult outside tutorials.

1. The Video Drills (Watch These First)

- **Scrimba Anchor 1:** Learn to Code with AI - Module: "Next-level projects with AI" *(Use this to understand how to leverage LLMs to generate complex Matplotlib boilerplate code)*[cite: 11, 13].
- **Scrimba Anchor 2:** Intro to AI Engineering - Module: "Your First AI Request" *(Baseline for understanding API request structures for automated data generation)*[cite: 12].
- **External Video 1:** YouTube: Python Pandas Tutorial - Working with Dates and Time Series Data *(Crucial for extracting datetime features dynamically)*.
- **External Video 2:** YouTube: Matplotlib Crash Course - Python Data Visualization *(Definitive guide to plotting data distributions and heatmaps)*.

2. Required Technical Documentation

- **Primary Architecture Doc:** Pandas Official Docs: Categorical data & One-Hot Encoding *(Required to convert strings into mathematical matrices)*.
- **Specific Syntax Doc:** Seaborn Official Docs: Plotting Correlation Heatmaps *(Required for your EDA deliverables)*.
- **Configuration Doc:** Jupyter Official Docs: Notebook Authoring & Best Practices *(Standard for local experimentation)*.

3. Engineering Assets

- **Data Asset:** Your Cleaned CSV from Day 1 *(You must ingest the flawless, machine-readable output you generated yesterday to proceed)*.

- **LLM Prompting Asset:** OpenAI Official Guide: Prompt Engineering Strategies *(Focus on prompting for correlation heatmaps and ROC-AUC curve scaffolding)*.
- **Configuration Asset:** GitHub Repository: Standard Jupyter .gitignore *(Prevents pushing local cache to production)*.

⚙️ Section B: The Execution Mandate (9:00 AM - 2:00 PM)

1. Execution Summary (TL;DR) Objective: Ingest yesterday's cleaned dataset to engineer advanced predictive features (datetime extraction, categorical encoding) and perform a comprehensive Exploratory Data Analysis (EDA) using Matplotlib and Seaborn.

2. The Workflow:

- **9:00 AM - 10:00 AM (Learning):** Watch the Matplotlib/Seaborn and Datetime drills. Read the Pandas documentation on categorical data encoding.
- **10:00 AM - 1:00 PM (Building):**
  - **Action 1 (Setup/Init):** Initialize a Jupyter Notebook strictly for initial experimentation and EDA. Import pandas, numpy, matplotlib.pyplot, and seaborn.
  - **Action 2 (Core Logic):** Dynamically extract new features from raw columns (e.g., extracting the day of the week, month, and holiday status from a datetime column). Convert string categories into mathematical integers using One-Hot Encoded matrices.
  - **Action 3 (Integration):** Generate professional EDA visualizations. You must output at least one Correlation Heatmap and two distribution plots tracking key target variables. Use an AI assistant (Cursor, Copilot, ChatGPT) to scaffold the complex Matplotlib boilerplate code.
- **The Non-Negotiable Rules:**
  1. **The ML Ban:** You are explicitly forbidden from importing Scikit-Learn, PyTorch, TensorFlow, or training any predictive models today. Today is strictly about feature engineering and data visualization.
  2. **The Interrogation Standard (Black Box Rule):** If you use Copilot/ChatGPT to generate Matplotlib scaling or distribution logic, you must be able to mathematically explain the operations occurring inside the "black box." Failure to articulate the theory results in immediate ticket rejection and penalized sprint velocity.
- **1:00 PM - 1:30 PM (Hygiene):** Run your notebook from top to bottom to ensure cell execution order is flawless. Export your final feature-engineered dataset as features_v1.csv. Push your .ipynb and CSV to GitHub.

- **1:30 PM - 2:00 PM (LinkedIn):** Draft a technical summary of your EDA. Document at least one statistical business insight derived from your correlation heatmaps. You must hyperlink your ProSensia admit card within the post text.

📥 Section C: Submission Protocol (Due strictly by 2:00 PM)

- **Step 1 (The Kanban Board):** Verify that this specific task card is shifted to the "Under Review" column on your team board.
- **Step 2 (The Assignments Module):** Navigate directly to today's entry in the platform's independent Assignment block. Paste both your public GitHub Repository URL and your live LinkedIn Post URL into the designated submission form fields.
- **Operational Note:** All grading and technical feedback loops operate through this portal. Do not DM Management your links.